Evaluation of Generative AI Outputs

ANL 310

Generative AI can draft, summarize, translate, classify, explain, and synthesize information, but its outputs require careful evaluation. In public health, generated content may affect case review, public communication, policy interpretation, community trust, staff training, or leadership decisions. Fluent writing does not guarantee accuracy, completeness, appropriateness, or safety. This module teaches learners how to evaluate generative AI outputs using structured review criteria. The module emphasizes factuality, source grounding, omissions, unsupported claims, bias, privacy, tone, accessibility, language quality, and fitness for intended use. Learners will distinguish low-risk drafting support from outputs that require stronger review because they could affect public health action. By the end of the module, learners should be able to create and apply an evaluation rubric for generative AI outputs in a public health workflow.

Level	intermediate/practitioner
Primary track	Analytics and Modeling Track
Appears in tracks	analytics-modeling

Learning Objectives

- Explain why generative AI outputs require structured review before public health use.
- Distinguish factuality, source grounding, completeness, relevance, tone, and accessibility.
- Identify hallucinations, omissions, unsupported claims, and misleading certainty.
- Design review criteria for different output types, including summaries, drafts, classifications, and recommendations.
- Apply stronger review requirements for sensitive, public-facing, or consequential outputs.
- Produce a generative AI output evaluation rubric.

Module Overview

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This module teaches learners how to evaluate generative AI outputs using structured review criteria. The module emphasizes factuality, source grounding, omissions, unsupported claims, bias, privacy, tone, accessibility, language quality, and fitness for intended use. Learners will distinguish low-risk drafting support from outputs that require stronger review because they could affect public health action.

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Why This Topic Matters for Public Health AI

Generative AI matters for public health because it can reduce burden in text-heavy workflows. Staff may use it to summarize guidance, draft meeting notes, simplify technical language, translate or adapt messages, organize case information, or synthesize stakeholder feedback. These uses can be valuable when they save time and improve clarity.

The same capabilities create risk. Generative AI can invent facts, omit caveats, misstate guidance, flatten uncertainty, or produce language that sounds more definitive than the evidence supports. It may introduce bias, inappropriate tone, or inaccessible wording. When sensitive data are involved, prompting and output handling can also create privacy and records concerns.

Public health agencies need structured evaluation because informal review is inconsistent. A reviewer may notice grammar errors but miss unsupported claims. Another may trust a polished summary without checking source documents. A rubric helps reviewers evaluate outputs against the purpose, source material, audience, and safeguards required for the use case.

Public Health Example

A communications team may use generative AI to draft a plain-language public message about a local respiratory illness increase. The draft may be clear and readable, but reviewers must check whether the numbers are correct, whether the recommendations match current guidance, whether uncertainty is represented accurately, and whether the message is accessible in multiple languages.

An epidemiology team may use generative AI to summarize case investigation notes for internal review. The summary must be checked against source notes because a missed exposure, incorrect date, or invented symptom could affect follow-up. The team should document whether the summary was accepted, edited, or rejected.

Technical Deep Dive

Generative AI evaluation should begin with intended use. A brainstormed outline, internal draft, public advisory, case summary, policy synthesis, or operational recommendation requires different levels of review. The more sensitive, public-facing, or consequential the use, the stronger the review should be.

Factuality review requires comparison to source material. Reviewers should check names, dates, numbers, conditions, legal requirements, recommendations, and causal statements. They should also verify that the output does not add unsupported claims or remove important caveats. For RAG systems, reviewers should inspect whether citations actually support the claims they accompany.

Completeness review asks whether the output includes the information needed for the task. A summary can be accurate but incomplete if it omits a critical exposure, eligibility criterion, limitation, or escalation instruction. Public health review should consider whether omitted information could change interpretation or action.

Equity and accessibility review should be included when outputs affect staff, partners, or the public. Reviewers should assess reading level, language access, disability accessibility, cultural context, stigma, blame, and whether the output addresses affected communities respectfully. Translation or multilingual adaptation should include review by qualified humans when used for public-facing or consequential communication.

Documentation should be proportionate to risk. Low-risk drafting may require only human review before use. Higher-risk workflows may require saved prompts, source references, reviewer initials, revision history, approval records, and incident escalation procedures.

Risks, Failure Modes, and Guardrails

Fluent but false outputs. A polished output may contain unsupported claims. Guardrails include source checking, citations, and reviewer signoff.

Omitted caveats. AI may simplify uncertainty away. Guardrails include completeness review and required caveat checks.

Inappropriate tone or accessibility. Outputs may be stigmatizing or inaccessible. Guardrails include plain language, equity, language access, and accessibility review.

Privacy exposure. Prompts or outputs may include sensitive data. Guardrails include approved tools, data-use tiers, redaction, access controls, and records guidance.

Application to AI-Supported Workflows

Generative AI output evaluation applies to drafting, summarization, classification, translation, code generation, and decision-support explanation. It is also relevant to RAG and agentic workflows because generated text may summarize retrieved sources or explain automated actions. Evaluation should be built into workflow design, not added after deployment.

The strongest approach is to create rubrics for specific output types. A case summary rubric will differ from a public message rubric, and a policy synthesis rubric will differ from a code review rubric. Each rubric should define what reviewers must check, who approves final use, and how errors are documented.

Reflection Questions

What type of generative AI output is being evaluated?

What source material should be used to verify the output?

What errors or omissions would be most harmful?

What review criteria are needed for equity, accessibility, language, and tone?

What documentation is required before the output can be used?

Practical Exercise

Create a generative AI output evaluation rubric for one public health use case. Include intended use, source verification, factuality, completeness, unsupported claims, omissions, bias, tone, accessibility, privacy, approval requirements, and escalation criteria.

Then apply the rubric to a sample AI-generated output and document whether the output should be accepted, edited, rejected, or escalated.

Expected Artifact or Evidence

Generative AI output evaluation rubric

Completed sample review

Source verification notes

Approval or escalation documentation

Expected Assignment

- Generative AI output evaluation rubric
- Completed sample review
- Source verification notes
- Approval or escalation documentation

Knowledge Check

- 1. Why is fluency not enough to trust a generative AI output?
- 2. What does grounding mean?
- 3. Which issue can be as harmful as an incorrect statement?
- 4. What should a rubric define?
- 5. Which output usually requires stronger review?

References and Resources

- NIST AI RMF Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- NIST Artificial Intelligence Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- OWASP Top 10 for Large Language Model Applications: https://owasp.org/www-project-top-10-for-large-language-model-applications/
- WHO Guidance on Large Multi-Modal Models: https://www.who.int/publications/i/item/9789240084759